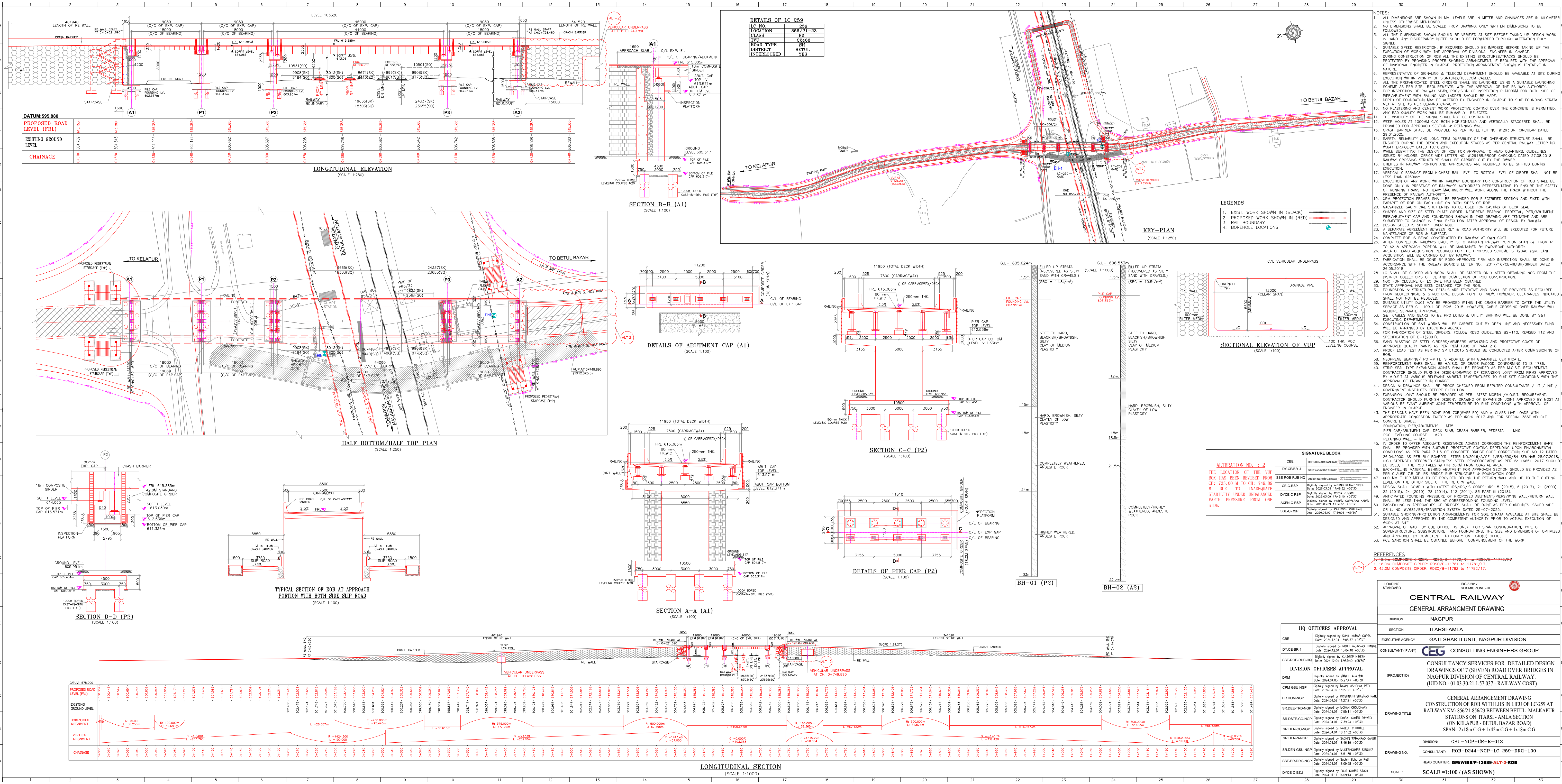




- NOTES:
1. ALL DIMENSIONS ARE SHOWN IN MM. LEVELS ARE IN METER AND CHAINAGES ARE IN KILOMETER UNLESS OTHERWISE MENTIONED.
 2. NO DIMENSIONS SHALL BE SCALED FROM DRAWING. ONLY WRITTEN DIMENSIONS TO BE FOLLOWED.
 3. ALL THE DIMENSIONS SHOWN SHOULD BE VERIFIED AT SITE BEFORE TAKING UP DESIGN WORK. IN HAND, ANY DISCREPANCY NOTED SHOULD BE FORWARDED THROUGH ALTERATION DULY NOTED.
 4. SUITABLE SPEED RESTRICTION, IF REQUIRED SHOULD BE IMPOSED BEFORE TAKING UP THE EXECUTION OF WORK WITH THE APPROVAL OF DIVISIONAL ENGINEER-IN-CHARGE.
 5. DURING CONSTRUCTION OF ROB ALL THE EXISTING STRUCTURES/TRACKS SHOULD BE PROTECTED BY PROVIDING PROPER SHORING ARRANGEMENT, IF REQUIRED WITH THE APPROVAL OF DIVISIONAL ENGINEER IN CHARGE. PROTECTION ARRANGEMENT SHOWN IS TENTATIVE IN NATURE.
 6. REPRESENTATIVE OF SIGNALING & TELECOM DEPARTMENT SHOULD BE AVAILABLE AT SITE DURING EXECUTION WITHIN VICINITY OF SIGNALING/TELECOM CABLES.
 7. ALL THE PREFABRICATED STEEL GIRDERS SHALL BE LAUNCHED USING A SUITABLE LAUNCHING SCHEME AS PER SITE REQUIREMENTS WITH THE APPROVAL OF THE RAILWAY AUTHORITY.
 8. FOR INSPECTION OF RAILWAY SPAN, PROVISION OF INSPECTION PLATFORM FOR BOTH SIDE OF PIER/ABUTMENT WITH RAILING AND LADDER SHOULD BE MADE.
 9. DEPTH OF FOUNDATION MAY BE ALTERED BY ENGINEER IN-CHARGE TO SUIT FOUNDING STRATA MET AT SITE AS PER BEARING CAPACITY.
 10. NO PLASTERING AND CEMENT WORK PROTECTIVE COATING OVER THE CONCRETE IS PERMITTED. ANY BAD QUALITY WORK WILL BE SUMMARILY REJECTED.
 11. THE VISIBILITY OF THE SIGNAL SHALL NOT BE OBSTRUCTED.
 12. WEET HOLES AT 100MM C/C BOTH HORIZONTALLY AND VERTICALLY STAGGERED SHALL BE PROVIDED FOR APPROACH SECTION & RETAINING WALL.
 13. CRASH BARRIER SHALL BE PROVIDED AS PER HQ LETTER NO. W.293.R. CIRCULAR DATED 29.01.2025.
 14. SAFETY, RELIABILITY AND LONG TERM DURABILITY OF THE OVERHEAD STRUCTURE SHALL BE ENSURED DURING THE DESIGN AND EXECUTION STAGES AS PER CENTRAL RAILWAY LETTER NO. W.641. BRPOLICY DATED 10.10.2018.
 15. WHILE SUBMITTING THE DESIGN OF ROB FOR APPROVAL TO HEAD QUARTERS, GUIDELINES ISSUED BY HQ/RS, OFFICE VIDE LETTER NO. W.2948R/PROOF CHECKING DATED 27.08.2018 RAILWAY CROSSING STRUCTURE SHALL BE CARRIED OUT BY THE OWNER.
 16. UTILITIES IN RAILWAY PORTION AND APPROACHES ARE REQUIRED TO BE SHIFTED DURING EXECUTION.
 17. VERTICAL CLEARANCE FROM HIGHEST RAIL LEVEL TO BOTTOM LEVEL OF GIRDERS SHALL NOT BE LESS THAN 6250mm.
 18. EXECUTION OF ANY WORK WITHIN RAILWAY BOUNDARY FOR CONSTRUCTION OF ROB SHALL BE DONE ONLY IN PRESENCE OF RAILWAY'S AUTHORIZED REPRESENTATIVE TO ENSURE THE SAFETY OF RUNNING TRAINS. NO HEAVY MACHINERY WILL WORK ALONG THE TRACK WITHOUT THE PRESENCE OF RAILWAY AUTHORITY.
 19. XPM PROTECTION FRAMES SHALL BE PROVIDED FOR ELECTRIFIED SECTION AND FIXED WITH PARAMET OF ROB ON EACH LINE ON BOTH SIDES OF ROB.
 20. GALVANIZED SACRIFICIAL SHUTTERING TO BE USED FOR CASTING OF DECK SLAB.
 21. SHAPES AND SIZE OF STEEL PLATE GIRDER, NEOPRENE BEARING, PEDESTAL, PIER/ABUTMENT, PIER/ABUTMENT CAP AND FOUNDATION SHOWN IN THIS DRAWING ARE TENTATIVE AND ARE SUBJECT TO CHANGE IN FINAL EXECUTION AFTER APPROVAL OF DESIGN BY RAILWAY.
 22. DESIGN SPEED IS 80kmph OVER ROB.
 23. A SEPARATE AGREEMENT BETWEEN RLY & ROAD AUTHORITY WILL BE EXECUTED FOR FUTURE MAINTENANCE OF ROB & SURFACE.
 24. AFTER COMPLETION RAILWAYS LIABILITY IS TO MAINTAIN RAILWAY PORTION SPAN i.e. FROM A1 TO A2 & APPROACH PORTION WILL BE MAINTAINED BY PMO/ROAD AUTHORITY.
 25. AREA OF LAND ACQUISITION REQUIRED FOR THE PROPOSED SCHEME IS 12040 sqm. LAND ACQUISITION WILL BE CARRIED OUT BY RAILWAY.
 26. FABRICATION SHALL BE DONE BY ROSS APPROVED FIRM AND INSPECTION SHALL BE DONE IN ACCORDANCE WITH THE RAILWAY BOARD'S LETTER NO. 2017/16/CE-III/BR/ORDER DATED 26.09.2016.
 27. CL SHALL BE CLOSED AND WORK SHALL BE STARTED ONLY AFTER OBTAINING NOC FROM THE DISTRICT COLLECTOR'S OFFICE AND COMPLETION OF ROB CONSTRUCTION.
 28. NOC FOR CLOSURE OF LC GATE HAS BEEN OBTAINED.
 29. STATE APPROVAL HAS BEEN OBTAINED FOR THE ROB.
 30. FOUNDATION & STRUCTURE DETAILS ARE TENTATIVE AND SHALL BE PROVIDED AS REQUIRED FROM GEOTECHNICAL & STRUCTURAL DESIGN POINT OF VIEW. HOWEVER, CLEARANCES INDICATED SHALL NOT BE REDUCED.
 31. SUITABLE UTILITY DUCT MAY BE PROVIDED WITHIN THE CRASH BARRIER TO CATER THE UTILITY SERVICE AS PER CL. 109.1 OF IRC-8:2015. HOWEVER, CABLE CROSSING OVER RAILWAY WILL REQUIRE SEPARATE APPROVAL.
 32. S&T CABLES AND GEARS TO BE PROTECTED & UTILITY SHIFTING WILL BE DONE BY S&T EXECUTING DEPARTMENT.
 33. CONSTRUCTION OF S&T WORKS WILL BE CARRIED OUT BY OPEN LINE AND NECESSARY FUND WILL BE ARRANGED BY EXECUTING AGENCY.
 34. FOR FABRICATION OF STEEL GIRDERS, FOLLOW ROSS GUIDELINES BS-110, REVISED 112 AND SPECIFICATION OF B-1(2011).
 35. REINFORCEMENT BARS SHALL BE A-15.2 OF GRADE Fe400, CONFORMING TO IS 1786.
 36. STRIP SEAL TYPE EXPANSION JOINTS SHALL BE PROVIDED AS PER M.O.S.T. REQUIREMENT.
 37. CONTRACTOR SHOULD FURNISH DESIGN/DRAWING OF EXPANSION JOINT APPROVED BY MOST AT VARIOUS RELEVANT AMBIENT TEMPERATURES TO SUIT SITE CONDITIONS WITH THE APPROVAL OF ENGINEER IN CHARGE.
 38. DESIGN & DRAWINGS SHALL BE PROOF CHECKED FROM REPUTED CONSULTANTS / IT / NT / GOVERNMENT INSTITUTES BEFORE EXECUTION.
 39. EXPANSION JOINT SHOULD BE PROVIDED AS PER LATEST MORTH / M.O.S.T. REQUIREMENT.
 40. CONTRACTOR SHOULD FURNISH DESIGN/DRAWING OF EXPANSION JOINT APPROVED BY MOST AT VARIOUS RELEVANT AMBIENT JOINT TEMPERATURE TO SUIT CONDITIONS WITH APPROVAL OF ENGINEER-IN-CHARGE.
 41. THE DESIGN HAS BEEN DONE FOR 70R(WHEELED) AND A-CLASS LIVE LOADS WITH APPROPRIATE CONGESTION FACTOR AS PER IRC-8:2017 AND FOR SPECIAL 380T VEHICLE.
 42. CONCRETE GRADE: FOUNDATION, PIER/ABUTMENTS - M35 PIER CAP/ABUTMENT CAP, DECK SLAB, CRASH BARRIER, PEDESTAL - M40 RETAINING WALL - M35.
 43. RETAINING WALL - M35.
 44. IN ORDER TO OFFER ADEQUATE RESISTANCE AGAINST CORROSION THE REINFORCEMENT BARS SHALL BE PROVIDED WITH SUITABLE PROTECTIVE COATING DEPENDING UPON ENVIRONMENTAL CONDITIONS AS PER PARA 7.1.5 OF CONCRETE BRIDGE CODE CORRECTION SLIP NO 12 DATED 26.04.2000. AS PER RLY BOARD'S LETTER NO.2016/4/CE-1/BR/350/84 SENARAY 28.07.2016. HIGH STRENGTH DEFORMED STEEL REINFORCEMENT AS PER IS: 16651-2017 SHOULD BE USED, IF THE ROB FALLS WITHIN 50M FROM COASTAL AREA.
 45. BACK-FILLING MATERIAL BEHIND ABUTMENT FOR APPROACH SECTION SHOULD BE PROVIDED AS PER CLAUSE 7.5 OF IRS BRIDGE SUB STRUCTURE & FOUNDATION CODE.
 46. 600 MM FILTER MEDIA TO BE PROVIDED BEHIND THE RETURN WALL AND UP TO THE CUTTING LEVEL ON THE OTHER SIDE OF THE RETURN WALL.
 47. DESIGN SHALL COMPLY WITH LATEST IRS/PC/S CODES: IRS: 5 (2015), 6 (2017), 21 (2000), 22 (2015), 24 (2010), 78 (2014), 112 (2011), 83 PART III (2018).
 48. ANTICIPATED FOUNDING PRESSURE OF PROPOSED ABUTMENT/PIERS/WING WALL/RETURN WALL SHALL BE LESS THAN THE SUB. AT CORRESPONDING SUBSTRATA LEVEL.
 49. BACKFILLING IN APPROACHES OF BRIDGES SHALL BE DONE AS PER GUIDELINES ISSUED VIDE CR. L. NO. W/881/BR/TRANSITION SYSTEM DATED 26-07-2025.
 50. SUITABLE SHORING/PROTECTION ARRANGEMENTS FOR SOIL STRATA AVAILABLE AT SITE SHALL BE DESIGNED AND APPROVED BY THE COMPETENT AUTHORITY PRIOR TO ACTUAL EXECUTION OF WORK AT SITE.
 51. APPROVAL OF GAD BY CBE OFFICE IS ONLY FOR SPAN CONFIGURATION, TYPE OF SUPERSTRUCTURE, SUBSTRUCTURE, AND FOUNDATIONS. THE SIZE AND DIMENSION OF OPTIMIZED AND APPROVED BY COMPETENT AUTHORITY ON C&CO OFFICE.
 52. PCE SANCTION SHALL BE OBTAINED BEFORE COMMENCEMENT OF THE WORK.

LEGENDS

1. EXIST. WORK SHOWN IN (BLACK)
2. PROPOSED WORK SHOWN IN (RED)
3. RAIL BOUNDARY
4. BOREHOLE LOCATIONS

SECTIONAL ELEVATION OF VUP (SCALE 1:100)

ALTERATION NO. : 2
THE LOCATION OF THE VUP BOX HAS BEEN REVISED FROM CH: 735.00 M TO CH: 749.89 M DUE TO INADEQUATE STABILITY UNDER UNBALANCED EARTH PRESSURE FROM ONE SIDE.

SIGNATURE BLOCK

CBE	DEEPAK NARAYAN KATE
DY-CE-BR-1	ROHIT VASWARO THAMRE
SSE-ROB-RUB-HQ	Asstt. North Circular
CE-CR-SP	Digitally signed by MRD KUMAR SINGH Date: 2024.03.09 17:43:10 +05'30'
DYCE-CR-SP	Digitally signed by RHEA KUMAR Date: 2024.03.09 17:43:10 +05'30'
AXEN-CR-SP	Digitally signed by VASWARO THAMRE KATE Date: 2024.03.09 17:43:10 +05'30'
SSE-CR-SP	Digitally signed by VASWARO THAMRE KATE Date: 2024.03.09 17:43:10 +05'30'

REFERENCES

1. 18.0M COMPOSITE GIRDER: RD50/B-11772/R1 to RD50/B-11772/R7
2. 22.0M COMPOSITE GIRDER: RD50/B-11781 to 11781/13.
3. 22.0M COMPOSITE GIRDER: RD50/B-11782 to 11782/17.

CENTRAL RAILWAY

GENERAL ARRANGEMENT DRAWING

DIVISION: NAGPUR

SECTION: ITARSI-AMLA

EXECUTIVE AGENCY: GATI SHAKTI UNIT, NAGPUR DIVISION

CONSULTANT (IF ANY): CEG CONSULTING ENGINEERS GROUP

CONSULTANCY SERVICES FOR DETAILED DESIGN DRAWINGS OF 7 (SEVEN) ROAD OVER BRIDGES IN NAGPUR DIVISION OF CENTRAL RAILWAY (UID NO: 01.05.30.21.1.57.037- RAILWAY COST)

DRAWING TITLE: GENERAL ARRANGEMENT DRAWING CONSTRUCTION OF ROB WITH LHS IN LIEU OF LC-259 AT RAILWAY KM: 85621-85623 BETWEEN BETUL-MALKAPUR STATIONS ON ITARSI - AMLA SECTION (ON KELAPUR - BETUL BAZAR ROAD) SPAN: 2x18m C.G + 1x42m C.G + 1x18m C.G

DIVISION: GBU - NGP - CR - R-042

CONSULTANT: ROB-D244-NGP-LC-259-DRG-100

HEAD QUARTER: GM/WB/P-13689-ALT-2-ROB

DRAWING NO.

SCALE: SCALE=1:100 (AS SHOWN)